How Well Can Agents Hunt Together? A QMIX Pursuit Study

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Agenda

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Motivation · Reinforcement Learning · Multi-Agent RL · The Pursuit Environment · MARL Challenges

Methodology

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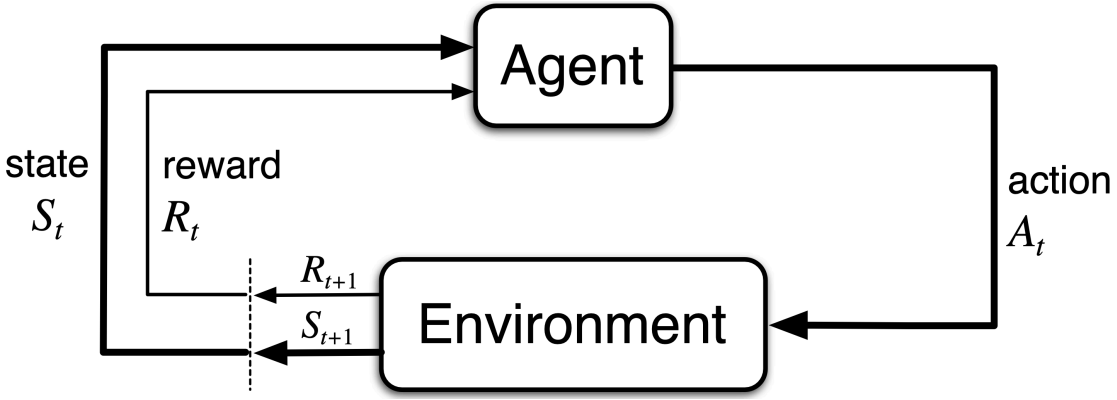
Motivation

Imagine a group of people trying to catch someone in a park.
One person alone has no chance but if they **spread out and cut off escape routes**, they succeed.

Nobody needs to talk. They just need to **read the situation and react together**.

How do we teach robots to do the same **purely from experience**?

Reinforcement Learning



- An **agent** observes the current **state** of its environment
- It selects an **action** → the environment transitions to a new state and returns a **reward**
- Goal: learn a **policy** π that maximises cumulative reward over time

> **Q-Value Q(s, a):** the expected total future reward of taking action *a* in state *s* — higher = better action.

Why Multi-Agent RL?

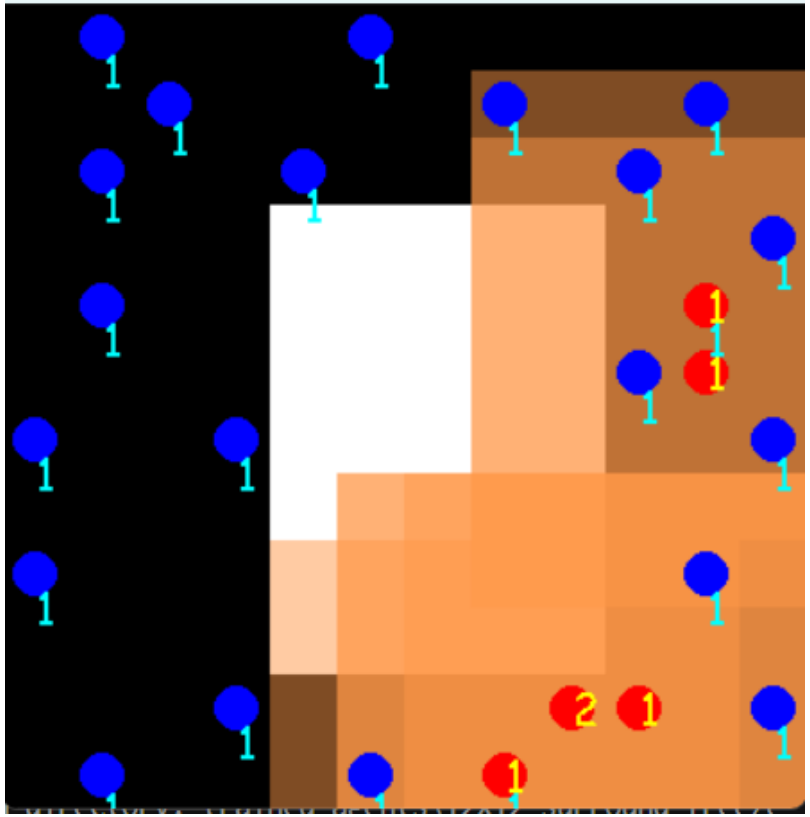
Many real-world problems require **multiple cooperating agents**:

- Robot swarms navigating a warehouse
- Autonomous vehicles merging at an intersection
- Search-and-rescue drones coordinating coverage

Single-agent RL assumes **one learner, one environment** — it cannot model cooperation or emergent team behaviour.

Multi-Agent RL (MARL) extends RL to settings where multiple agents learn simultaneously and must coordinate.

The Pursuit Environment



- Red — Pursuers — the agents you control
- Blue — Evaders — the targets to catch
- White — Obstacles — walls blocking movement

The Pursuit Environment

- Discrete grid: 8×8 and 12×12
- Each pursuer sees a local 7×7 patch
- 5 actions: Up, Down, Left, Right, Stay

	Freeze	Active
Touch	Contact triggers capture	Evader escapes actively
Surround	Must fully enclose evader	Must enclose a moving target

Reward: +5 capture · +0.01 tag · −0.1 per step

> The reward structure is defined by PettingZoo. The logic: +5 incentivises cooperation towards capture, +0.01 discourages passive waiting, −0.1/step keeps episodes short and purposeful.

MARL Challenge 1: Non-Stationarity

In single-agent RL, the environment is a **fixed MDP** — convergence relies on this.

In MARL, every agent's policy is **changing simultaneously**:

- Other agents are part of the environment
- But their behavior keeps shifting as they learn
- The MDP is **non-stationary** from every agent's perspective

> Standard Q-learning convergence guarantees break down entirely.

MARL Challenge 2: How Do We Give Rewards?

Option A — Individual rewards: each agent gets its own reward based on what it did personally.

- Agents train independently, only optimizing for themselves
- Agent 3 learns to chase captures solo — it ignores teammates entirely
- Agents may even **compete** for the same evader instead of coordinating
- → **Selfish behavior, no teamwork**

Option B — One shared group reward: the whole team gets the same reward after every step.

- Forces agents to care about the team outcome
- → **Cooperation emerges** — but now a new problem appears...

MARL Challenge 3: Credit Assignment

The team captured an evader. Everyone gets +5. But:

- Agent 1 moved into the perfect flanking position ✓
- Agent 2 made the final contact ✓
- Agent 3 wandered in the corner doing nothing ✗

All three get the same reward. The learning signal cannot distinguish good from bad behavior.

> Which agent actually deserved the reward?
> How does the system know **who to train more** and **who to correct?**

This is the **credit assignment problem** — and it is what QMIX is designed to solve.

QMIX: The Big Picture

```
Local obs τ¹  →  Agent Network  ──→  Q¹  ┐
Local obs τ²  →  Agent Network  ──→  Q²  ├──→  Mixing Network  ──→  Q_tot  ──→  Loss
Local obs τⁿ  →  Agent Network  ──→  Qⁿ  ┘
                                                      ↑
                              Global state s  ──→  Hypernetworks  ──→  W, b
```

- Each agent produces **one Q-value** for the action it took
- The **mixer** combines them into a single team value Q_tot
- The **hypernetworks** read the global state and generate the mixer's weights

> **Q-value reminder:** Q(s, a) = how good is action *a* in state *s*. Here each agent outputs one Q-value for its chosen action, and the mixer learns to weight them into a team score.

Agent Networks

Each pursuer runs the **same shared network** (parameter sharing):

```
Local observation (147 values)
        ↓  fc1 → ReLU
        ↓  fc2 → ReLU
        ↓  fc3
  [Q(Up), Q(Down), Q(Left), Q(Right), Q(Stay)]
```

- Input: what the agent currently sees, flattened to **147 numbers**
- Output: **5 scores**, one per possible action
- The agent picks the action with the **highest score**
- That one winning score gets passed to the mixer

> Two agents see different things → different scores → different actions, even from the same network.

⚠️ **In this project:** all agents share **one single network**

Hypernetworks: State-Dependent Weights

The mixing network has **no fixed weights** — they are generated fresh every step.

Two hypernetworks take the global state s (all agent + evader positions):

$$s \xrightarrow{\text{hyper_w1}} W_1 \qquad s \xrightarrow{\text{hyper_w2}} W_2$$

$$s \xrightarrow{\text{hyper_b1}} b_1 \qquad s \xrightarrow{\text{hyper_b2}} b_2$$

The mixing computation is then:

$$\text{hidden} = \text{ELU}([Q^1 \dots Q^n] \cdot |W_1| + b_1)$$

$$Q_{\text{tot}} = \text{hidden} \cdot |W_2| + b_2$$

Monotonicity: Why |·| Matters

The mixing weights W1 and W2 are passed through absolute value → always non-negative.

Formal condition:

$$\frac{\partial Q_{\text{tot}}}{\partial Q^a} \geq 0 \quad \forall a \in \{1, \ldots, n\}$$

This means: if any agent improves its individual value, the team value can only increase.

Why this enables decentralized execution:

$$\arg\max_{\mathbf{u}} Q_{\text{tot}}(s, \mathbf{u}) = \begin{pmatrix} \arg\max_{u^1} Q^1 \\ \vdots \\ \arg\max_{u^n} Q^n \end{pmatrix}$$

Each agent picks its own best action independently — **no communication or central controller needed at runtime**.

How Training Works End-to-End

One forward pass:

1. Each agent network produces Q^a from local observation
2. Hypernetworks read global state → generate W_1, W_2, b_1, b_2
3. Mixer computes Q_{tot}
4. Loss: $\mathcal{L} = \left(r + \gamma Q_{\text{tot}}^{\text{target}} - Q_{\text{tot}}\right)^2$

One backward pass — one big connected network:

- Loss flows back through the mixer into every agent network

> The mixer decides how much credit each agent receives — this is how QMIX solves credit assignment.

Centralized Training, Decentralized Execution

	Training	Execution
Global state	✅ Available to hypernetworks	❌ Discarded
Mixer + hypernetworks	✅ Used for Q_tot	❌ Discarded
Agent network	✅ Updated via backprop	✅ Used alone
Communication	❌ Not used	❌ Not used

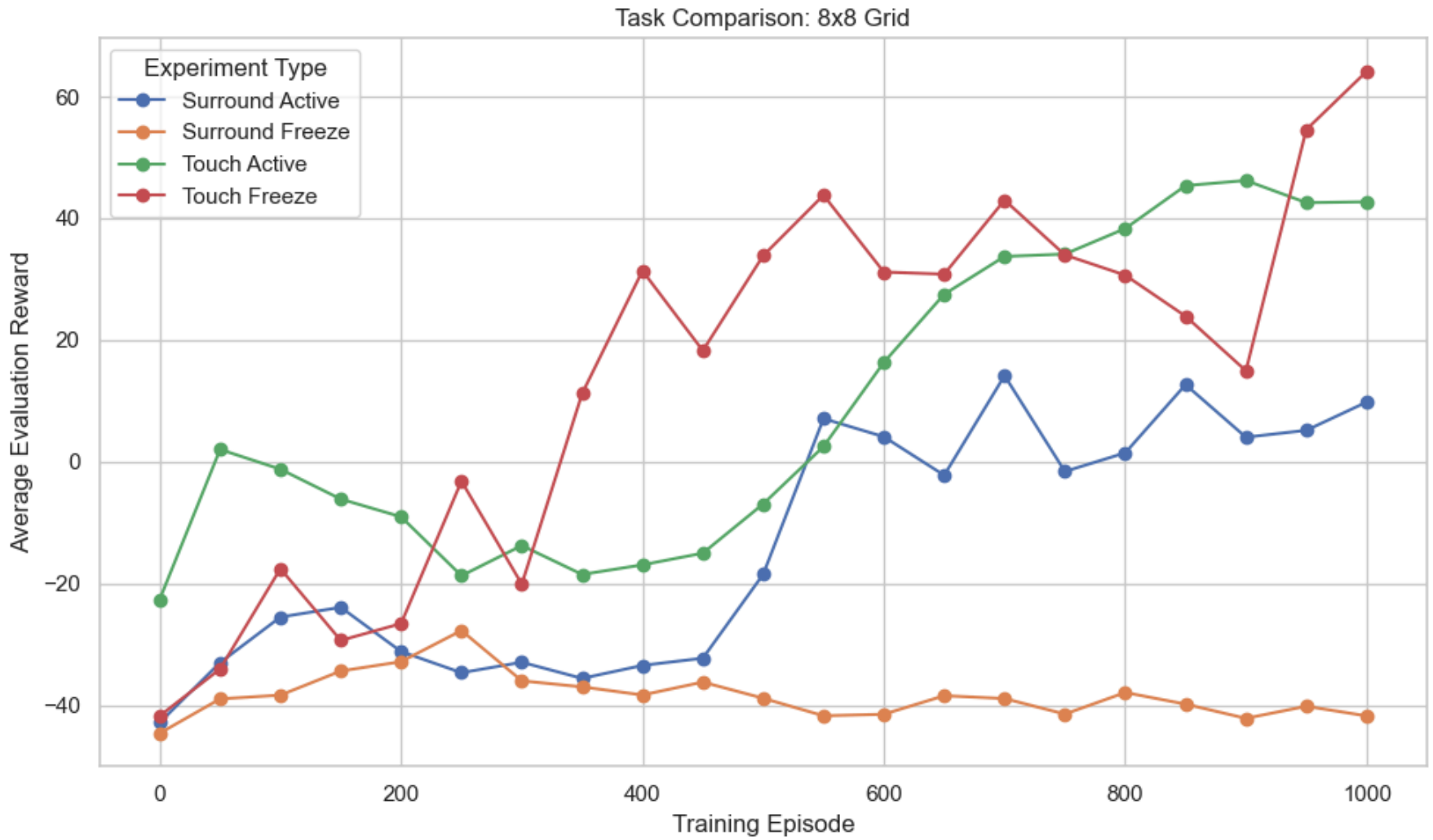
At deployment: only the agent network runs. No mixer, no global state, no communication.
Coordination is entirely implicit — baked into the Q-values during training.

Experimental Setup

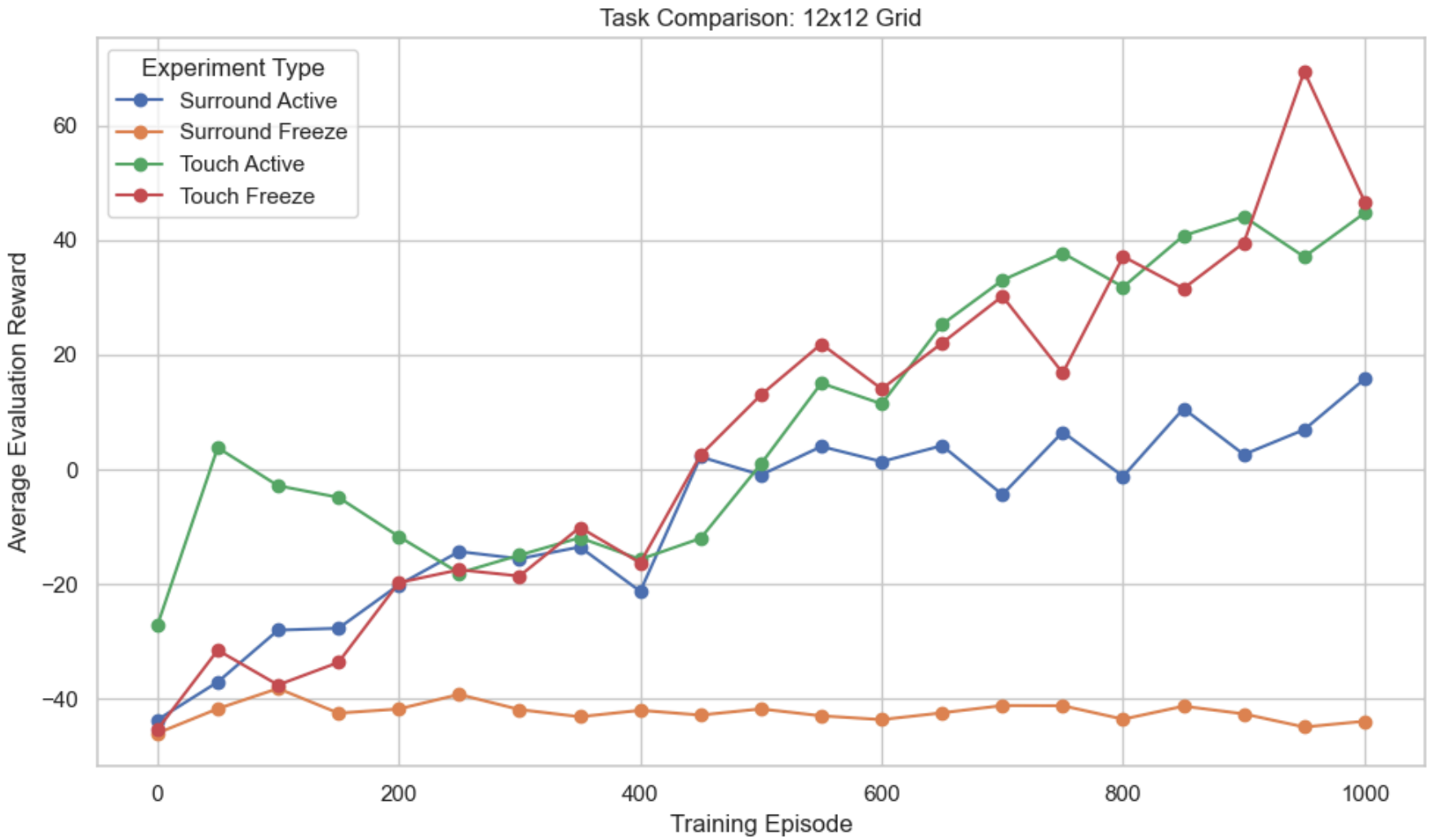
- **8 experiments:** 2 grid sizes × 4 task variants
- 4 pursuers on 8×8 · 6 pursuers on 12×12

Evaluation protocol: every 50 episodes, pause training → run 10 evaluation episodes → log average reward

Results: Task Comparison (8×8)



Results: Task Comparison (12×12)



Results: Best Rewards

Strategy	8×8	12×12
Surround Freeze	−27.68	−38.17
Surround Active	+14.18	+15.79
Touch Freeze	**+64.13**	**+69.43**
Touch Active	**+46.22**	**+44.73**

Capture condition matters far more than grid size.

Surround → Touch: **+30 to +50 points**. Surround/Freeze completely fails to learn.

Why Surround/Freeze Fails

- Surround requires **simultaneous enclosure** — hard joint action
- Freeze = **static environment** → agents see the same observations repeatedly

Infinite loop problem:

1. Agents revisit the same state → same action → no capture
2. No negative signal → nothing forces them to change
3. Even the best policy gets stuck if the environment never changes

The active evader **breaks this** — a moving target means the same action sequence stops working, forcing adaptation.

Why Touch/Freeze Performs Way Better

Agents often start near each other — and near evaders.

- They quickly discover that **moving in pairs and touching the same evader** gets a reward
- The strategy is simple enough to stumble upon early in training
- Even in a static environment, the **touch pattern is easy to repeat and reinforce**

> agents find a working solution before they could get stuck.

Scaling Results

Future Work

Direction	Problem it solves
GRU (Gated Recurrent Unit) agent network	Agents forget evaders leaving the 7×7 window — recurrent memory would help
Communication & roles	Agents coordinate implicitly — explicit messages could enable role differentiation → fix Surround failure)

Sources

	Reference
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[5]	Sunehag et al. (2018). *Value-Decomposition Networks For Cooperative Multi-Agent Learning*. AAMAS.

Thank You